Sohan Jayram Reddy Kadiri

Seattle | sohanjreddy@outlook.com | (206) 380-6390 | github.com/sohanreddyk | linkedin.com/in/sohanreddyk | www.sohanreddy.org/

PROFESSIONAL SUMMARY

Software Engineer with experience building full-stack applications and distributed backend services using Java, Python, TypeScript and Go. Expertise in API development, performance optimization and cloud infrastructure.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Engineered 2 **full-stack applications using React, Node.js and PostgreSQL**, integrating REST APIs and authentication to serve 2,500+ users.
- Improved AI chatbot retrieval relevance by 40% through a **Pinecone-backed RAG pipeline**, supporting 1,200+ conversations with 850 ms response latency.
- **Led 3 interns** through architecture design and API integration, resolving 9 production issues through systematic debugging and service-level testing.
- **Automated** frontend and backend deployments using **Docker, Azure, and Vercel CI/CD pipelines**, reducing release time by 70%, from 20 to 6 minutes.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Developed a **full-stack digital marketplace** serving 8,000+ users and processing 1,600+ orders, generating $20K in gross merchandise value.
- Increased checkout conversion from 41% to 50% (**22% relative improvement**) through **funnel optimization and one-click checkout**, contributing $14K in revenue.
- Engineered a real-time messaging system using **WebSockets**, supporting 350 concurrent connections with **sub-150 ms** latency and 99.8% message delivery.
- Improved mobile conversion by 8% through **Lighthouse-guided** optimizations, reducing page load time to under 1.2 seconds.

PROJECTS

Insight — Real-Time Product Analytics Platform | TypeScript, NestJS, Kafka, PostgreSQL, Redis, Next.js, Docker

- Engineered a **multi-tenant product analytics platform** using NestJS, Kafka and PostgreSQL to ingest user events and deliver real-time dashboards through server-sent events.
- Implemented **idempotent Kafka processing**, transactional PostgreSQL aggregation and dead-letter queues to prevent duplicate event counts and recover from processing failures.
- Increased event-processing throughput by **61×** through batched PostgreSQL writes and bulk aggregation, sustaining 12,000 events/sec at 180 ms p95 ingestion latency.

InspectIQ — AI-Assisted Vehicle Inspection Platform | TypeScript, React, Python, PostgreSQL, AWS, Terraform

- Developed a **cloud-native vehicle inspection platform** integrating web and mobile workflows, role-based authentication and AI-assisted image analysis.
- Designed an asynchronous **S3–SQS–Lambda pipeline** with structured AI-output validation, human approval and auditable inspection reports.
- Automated **API, integration and browser testing** through CI, achieving 92% test coverage and processing 1,000 images/hour at 2.8 s p95 latency.

TaskForge — Distributed Task Queue | C++17, Redis, Multithreading, Python, Next.js, Docker

- Engineered a **distributed task queue in C++17 and Redis** with multithreaded workers, priority scheduling and asynchronous execution, targeting 5,000 tasks/sec.
- Implemented **at-least-once delivery, configurable retries and stuck-task recovery**, with 99.9% task completion and recovery within 5 seconds during failure testing.
- Optimized **concurrent task execution and worker scheduling**, targeting 20 ms p95 dispatch latency across 16 concurrent workers.

SKILLS

Languages: Java, Python, C++, Go, TypeScript, JavaScript, SQL

Frameworks: React, Next.js, Node.js, NestJS, FastAPI, Spring Boot

Databases & Messaging: PostgreSQL, Redis, MongoDB, Kafka, RabbitMQ

Cloud & Tools: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Prometheus, Grafana

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025